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JPMethod Polynomials & Quadratics

Algebraic Expressions

Polynomials & Quadratic Equations

Algebraic Expressions

This workbook teaches polynomial manipulation and quadratic equation solving through progressive exercises.

- Start with polynomial multiplication
- Learn multiplication using algebraic formulas
- Master factorisation techniques
- Work with square roots and quadratic equations
- Explore graphs of quadratic functions
- Apply the Pythagorean theorem

Regular practice with polynomials builds strong algebraic foundations.

Section 1. Multiplication of Polynomials

Multiply the polynomials and simplify your answer.

(1) $2a(2a + 4) =$

(2) $2a(3a + 9) =$

(3) $(x + 1)(x + 1) =$

(4) $3a(3a - 9) =$

(5) $5a(2a + 1) =$

(6) $(y + 3)(y - 2) =$

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Guide for Parents and Tutors

This JPMMethod Polynomials & Quadratics workbook provides structured practice for Algebraic Expressions.

How to Use This Workbook

- Work through sections in order
- Complete practice problems before checking solutions
- Review mistakes to improve understanding
- Practice regularly for best results

Understanding Polynomial Operations

This workbook covers fundamental polynomial operations including multiplication, factoring, and solving quadratic equations. Each section builds on previous concepts.

Common Challenges

- Students may confuse distribution with simple multiplication
- Sign errors are common when working with negative terms
- Factoring requires pattern recognition skills
- Quadratic formula needs careful substitution

Support Strategies

Encourage students to show all work and check answers by substitution. Visual aids and graph paper can help with understanding polynomial behavior.

Section 1. Multiplication of Polynomials

Multiply the polynomials and simplify your answer.

$$(1) \quad 2y(1y - 6) =$$

$$(2) \quad (x + 4)(x - 5) =$$

$$(3) \quad 4a(3a + 6) =$$

$$(4) \quad 2a(1a + 5) =$$

$$(5) \quad 2y(2y - 8) =$$

$$(6) \quad 4y(1y + 5) =$$

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Section 1. Multiplication of Polynomials

Multiply the polynomials and simplify your answer.

(1) $3x(2x - 7) =$

(2) $4x(1x - 1) =$

(3) $(y - 1)(y + 2) =$

(4) $(y - 4)(y + 2) =$

(5) $4a(3a - 7) =$

(6) $(x - 2)(x + 5) =$

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Section 1 - Solutions

1a-1: $4a^2 + 8a$

1a-2: $6a^2 + 18a$

1a-3: $x^2 + 2x + 1$

1a-4: $y^2 - 2y - 8$

1a-5: $10a^2 + 5a$

1a-6: $y^2 + 1y - 6$

1b-1: $4a^2 + 8a$

1b-2: $x^2 - 1x - 20$

1b-3: $12a^2 + 24a$

1b-4: $9a^2 - 27a$

1b-5: $10a^2 + 5a$

1b-6: $4y^2 + 20y$